THE UNIVERSITY OF AZAD JAMMU AND KASHMIR MUZAFFARABAD



IMAGE PROCESSING TECHNOLOGY

Lab Instructor: Engr. Mam Sidra
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Lab Task: 3

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1. Introduction

This practical is based on the final four tasks given at the end of Lab 03. The tasks use NumPy to perform basic calculations and OpenCV to create an image, write a custom message on it, change its dimensions, and display red text.

2. Task I — Change the NumPy Array

Requirement: Change the numbers array to [5, 10, 15] and calculate its sum and average.

Python code:

```
import numpy as np

numbers = np.array([5, 10, 15])

print("Numbers:", numbers)
print("Sum:", np.sum(numbers))
print("Average:", np.mean(numbers))
```

Output:

```
Numbers: [ 5 10 15]
Sum: 30
Average: 10.0
```

Explanation: The sum is 5 + 10 + 15 = 30. The average is 30 divided by 3, which equals 10.0.

3. Task II — Change the OpenCV Text

Requirement: Replace the sample text “Hello OpenCV!” with a personal message.

The message used in this solution is: “My name is Jaffar!”

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

image = np.zeros((200, 500, 3), dtype=np.uint8)

cv2.putText(
    image,
    "My name is Jaffar!",
    (35, 110),
    cv2.FONT_HERSHEY_SIMPLEX,
```

```
    1,
    (0, 0, 255),
    2
)

cv2_imshow(image)
```



4. Task III — Change Image Dimensions

Requirement: Change the image dimensions from (300, 400, 3) to (200, 500, 3).

```
image = np.zeros((200, 500, 3), dtype=np.uint8)

print("Image shape:", image.shape)
```

Output:

```
Image shape: (200, 500, 3)
```

Explanation: The image has a height of 200 pixels, a width of 500 pixels, and 3 colour channels. The data type uint8 stores each channel value from 0 to 255.

5. Task IV — Use Red Text Colour

Requirement: Use the OpenCV colour value (0, 0, 255) for red text.

OpenCV uses BGR colour order. Therefore, (0, 0, 255) represents red (Blue = 0, Green = 0, Red = 255).

```
import cv2
import numpy as np
```

```
from google.colab.patches import cv2_imshow

image = np.zeros((200, 500, 3), dtype=np.uint8)

cv2.putText(
    image,
    "My name is Jaffar!",
    (35, 110),
    cv2.FONT_HERSHEY_SIMPLEX,
    1,
    (0, 0, 255),
    2
)

cv2_imshow(image)
```

Output image:



6. Complete Combined Program

```
import numpy as np
import cv2
from google.colab.patches import cv2_imshow

# Task I: NumPy array, sum and average
numbers = np.array([5, 10, 15])
print("Numbers:", numbers)
print("Sum:", np.sum(numbers))
print("Average:", np.mean(numbers))

# Tasks II, III and IV: OpenCV image and red text
image = np.zeros((200, 500, 3), dtype=np.uint8)
```

```
cv2.putText(
    image,
    "My name is Jaffar!",
    (35, 110),
    cv2.FONT_HERSHEY_SIMPLEX,
    1,
    (0, 0, 255),
    2
)

print("Image shape:", image.shape)
cv2_imshow(image)
```

```
Numbers: [ 5 10 15]
Sum: 30
Average: 10.0
Image shape: (200, 500, 3)
```



7. Conclusion

All four tasks from the end of Lab 03 have been completed. The NumPy array was changed and its sum and average were calculated. The OpenCV sample text was replaced with a personal message, the image dimensions were changed to (200, 500, 3), and the text colour was set to red using the BGR value (0, 0, 255).